

A systematic literature review on sugarcane crop yield and disease prediction using deep learning and remote sensing

ABSTRACT

Sugarcane is an important cash crop and a significant source of bioethanol and sugar globally. Stable production of sugarcane demands reliable yield estimation and disease detection at an early stage, both of which are important for enhancing food security as well as for ensuring sustainable agriculture. Conventional field- and statistical methods of tracking sugarcane yield and health are usually spatially confined, labor-intensive, and incapable of exploring intricate interactions among environmental, climatic, and biological variables. The development of remote sensing and deep learning has introduced new possibilities for confronting these limitations using scalable, data-driven methods.

In this paper, a systematic literature review of deep learning-based sugarcane yield and disease forecasting research using remote sensing data is put forward. Research studies from the years 2012 to 2024 were retrieved from prominent databases and examined thoroughly. Triggered and conditioning factors, datasets and observation time, disease types, factor elimination methods, algorithms, and performance metrics are emphasized in the review. Our analysis reveals that convolutional neural networks (CNN), long short-term memory (LSTM) models, and their hybrids are the most commonly used deep learning techniques, and vegetation indices from multispectral and hyperspectral imagery continue to be the most prevalent features for prediction. Yet, major limitations still exist in the form of small or region-specific datasets, heterogeneous feature selection approaches, and limited generalization across disease types and agro-climatic zones.

By synthesizing results across a range of studies, this Systematic literature review specifies major research gaps and directions. Specifically, the integration of multi-source remote sensing data, the use of interpretable deep learning architectures, and the augmentation of disease-specific datasets are the key towards developing strong, scalable, and actionable prediction frameworks. The findings from this review aim to assist researchers, farm practitioners, and policymakers in driving precision agriculture for sugarcane.

1. INTRODUCTION

Sugarcane (*Saccharum officinarum* L.) is one of the most important commercial crops in the world, grown mainly in tropical and subtropical areas for sugar, ethanol, and bioenergy production. It accounts for close to 80% of the world's sugar production and supports the economies of many developing countries, most notably India and Brazil, where millions of farmers rely on it for a living, and industries like sugar mills, ethanol distilleries, and cogeneration plants depend on it as a raw material supply [1,2]. Stable sugarcane production is thus important for ensuring food security, industrial planning, and policy-making.

Sugarcane productivity and growth are extremely sensitive to environmental, biological, and management factors. Climatic fluctuations like rainfall, temperature, and humidity tend to have a direct influence on phenological phases and yields [3,4]. Crop health is also strongly conditioned by soil fertility and irrigation management. Conversely, diseases including red rot, smut, rust, and mosaic virus are the most destructive biotic stresses, causing tremendous yield losses in case they are not detected and controlled at an early point [5]. Conventional field inspection and laboratory

tests for disease monitoring are time-consuming, labor-intensive, and offer late feedback, which restricts their potential for real-time action.

The aggressive advancement of remote sensing technology has made it possible to monitor sugarcane on large scales at various spatial and temporal resolutions. Satellite sensors (e.g., MODIS, Landsat, Sentinel), UAV-mounted multispectral and hyperspectral imaging, and ground sensors are being used more and more to monitor crop dynamics [6]. Vegetation indices like NDVI, EVI, and SAVI computed from these data have been related to sugarcane biomass, phenological phases, and productivity [7,8]. UAVs, while expensive and with limited coverage, offer high-resolution field-scale observations of stress factors.

In parallel to these developments, artificial intelligence (AI) and deep learning have been leading-edge tools in predictive agriculture. Early machine learning techniques like Random Forests, SVMs, and ANNs showed promise for yield and disease prediction [9], but their reliance on handcrafted features made them unscalable. Deep learning algorithms, i.e., Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) like LSTMs and GRUs, are now extensively applied due to their capability to learn automatically spatial and temporal representations from unprocessed data [10,11]. CNNs are best in extracting spatial features from images, and LSTMs are good at extracting temporal relationships from climatic and phenological time series. Later work includes attention mechanisms, Graph Neural Networks (GNNs), and multi-source data fusion to enhance prediction robustness [12–14].

Notably, sugarcane disease and yield are symbiotically connected. Disease incidence directly limits productivity, and therefore examining them in isolation may miss valuable interactions. Coupled models that integrate disease severity indices with yield prediction frameworks have achieved as much as 15% increase in accuracy under biotic stress [15]. These measures are, however, few in number due to difficulties in matching datasets, prescribing multi-task loss functions, and achieving generalizability across varied agro-climatic zones.

In spite of these developments, a number of limitations remain. Oftentimes, small, regional datasets are used, which limits model transferability. Deep learning models are also computationally demanding, which necessitates the use of high-performance GPUs, limiting deployment on UAVs or mobile devices in actual farm environments. In addition, their "black-box" property detracts from the confidence of farmers and practitioners, so it is crucial for explainable AI (XAI) methods like Grad-CAM and SHAP to be employed [16].

Considering these limitations, an SLR is mandatory to summarize the current body of evidence, assess methodological merits and flaws, and highlight gaps in research on sugarcane yield and disease prediction using deep learning and remote sensing. The review covers initiated factors (e.g., the occurrence of disease, pest episodes) and conditioning factors (e.g., climatic factors, soil type, management practices), datasets and observation windows, algorithms, feature selection strategies, and performance measures. It also evaluates the secondary study quality, suggests a systematic agenda for integrated models, and summarizes implications for policymakers, practitioners, and researchers.

The rest of the paper is structured as follows: Section 2 describes the study selection process based on PRISMA and methodology. Section 3 summaries yield prediction, disease detection, and integrated approaches and provides their critical appraisal. Section 4 discusses results, limitations, and directions for future research, and Section 5 concludes with practical implications for various stakeholders.

2. RESEARCH METHODOLOGY

The research employs a Systematic Literature Review (SLR) to synthesize and analyze the available body of literature on sugarcane yield and disease forecasting through deep learning and remote sensing. The evidence-based guidelines of Kitchenham and Charters (2007) [17] are followed in the methodology, and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol [18] is followed. These frameworks were chosen since they guarantee transparency, reproducibility, and rigor in the identification, evaluation, and synthesis of past research. Applying these protocols systemically, the review not only gives a critical summary of the research situation as it stands at present but also guarantees that the conclusions reached are founded on well-specified, reproducible procedures instead of subjective opinion.

2.1 Research Questions

In the beginning, a set of research questions (RQs) were created to inform the review. The initial question (RQ1) asked what deep learning methods have been utilized for sugarcane yield prediction and disease identification, ranging from traditional to cutting-edge architectures. The second research question (RQ2) concerned the remote sensing technologies and data sources employed in the studies, because various sensors (e.g., satellite, UAV, hyperspectral, multispectral) offer different resolutions that impact model performance. The third research question (RQ3) queried the degree to which studies have made use of triggered factors (e.g., disease incidence, pest outbreaks, or biotic stress events) and conditioning factors (e.g., climate, soil, and management practices) in their models. The fourth question (RQ4) explored the datasets themselves, examining size, temporal coverage, spatial scope, and disease categories. The fifth question (RQ5) searched the reported benefits, limitations, and uses of various approaches, and the last question (RQ6) aimed at determining research gaps and formulating a forward agenda. Together, these queries guaranteed that the review would go beyond a mere listing of research and instead provide a guided and critical synthesis.

2.2 Search Strategy

For maximizing coverage, a thorough search strategy was used in six well-known digital libraries: Scopus, Web of Science, IEEE Xplore, ScienceDirect, MDPI, and Google Scholar. Keywords were combined with Boolean operators for comprehensive capture of a broad array of relevant literature. Common search terms were phrases like "Sugarcane yield prediction AND deep learning AND remote sensing", "Sugarcane disease detection OR disease prediction AND deep learning", and "Crop yield prediction AND sugarcane AND UAV OR satellite". These terms were formulated to include both yield and disease considerations, and to reflect the expanding use of UAVs and multi-source remote sensing in crop prediction [19].

Google Scholar, while popular, poses a special challenge due to its algorithmic and non-transparency in indexing. In an effort to avoid bias, this review adhered to a systematic rule where the first 200 most relevant results were screened. This was accompanied by a snowballing approach where references of the already identified studies were further searched to identify other relevant works. The synergy of formal database querying and citation expansion based on references guaranteed that core and peripheral literature were both encompassed within the review corpus [20].

2.3 Inclusion and Exclusion Criteria

To narrow down the list of studies, strict inclusion and exclusion criteria were used. Only publications between 2012 and 2024 were included, as this window of time marks the beginning and explosive

rise of deep learning within agricultural science [21]. Research had to explicitly focus on sugarcane yield or disease forecasting and use deep learning techniques like CNNs, LSTMs, hybrids, attention-based models, or Graph Neural Networks. Additionally, included research had to work on remote sensing data, either from satellite, UAV, or hyperspectral/multispectral images, so that the emphasis was on data-driven and scalable approaches. Publications were restricted to English-language peer-reviewed journal articles and conference papers, and every study had to provide quantitative performance measures such as RMSE, R^2 , accuracy, or F1-score [22,23].

Exclusions were also critical. Articles dealing with general crop yield prediction without applying to sugarcane were excluded. In the same way, articles that had used only statistical or traditional machine learning methods, without deep learning, were excluded to ensure thematic coherence [24]. Articles not published in English, duplicate entries from more than one database, full-text publications not accessible, and grey literature (theses, white papers, and technical reports not undergoing peer review) were removed systematically. These standards ensured that the end corpus was valid and relevant, as well as reliable, for evidence-based synthesis.

2.4 Study Selection Process

The process of article selection was conducted following the four-step PRISMA protocol of identification, screening, eligibility, and inclusion. At the identification stage, 1,190 articles were yielded across the databases. In the screening stage, duplicate entries were eliminated and titles and abstracts were reviewed to rule out non-relevant studies, cutting down the pool significantly. At the eligibility phase, full-text articles were examined in detail against inclusion and exclusion criteria. Some were excluded here for not having specific sugarcane focus, using solely traditional machine learning, or insufficient methodological information. Lastly, at the inclusion phase, 61 studies remained for detailed synthesis and analysis.

This multi-step procedure is illustrated in a PRISMA flow diagram (Figure 1), which summarizes the number of identified records, screened, excluded, and included at each step, together with the reasons for exclusion. Through this systematic method, the study maximizes transparency on how the included final set of reviewed papers was arrived at, ensuring increased reproducibility and credibility of the review [18].

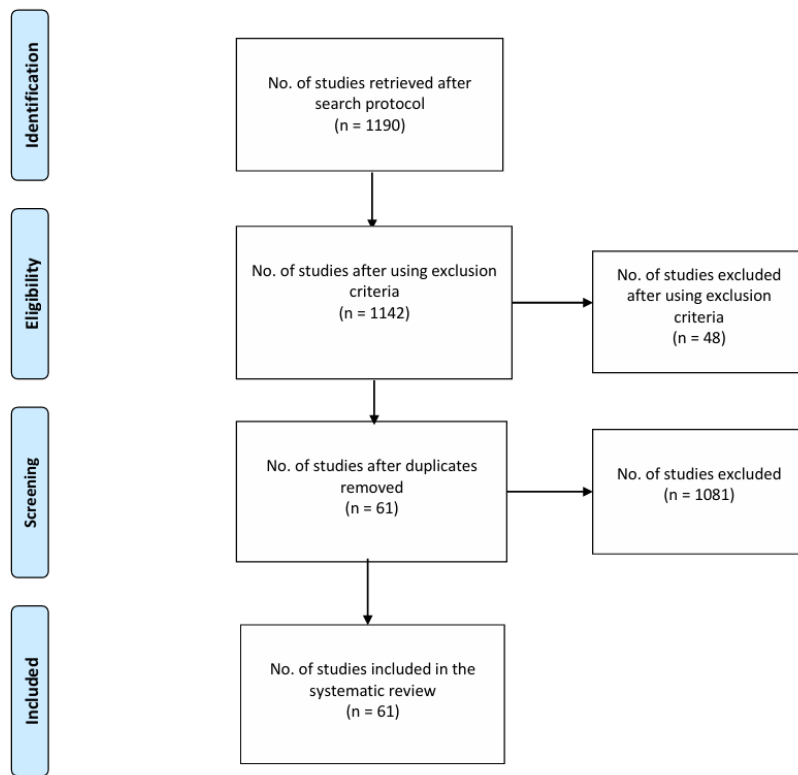


Figure1. PRISMA flowchart showing the results of searches.

Table1. Distribution of articles among various databases.

Databases	No. of retrived studies after search protocol	No. of studies after exclusion criteria	No. of studies after removing repeated articles
Science Direct	716	33	683
ACM	122	3	119
IEEE Xplore	346	20	326
MDPI	6	5	1
Total	1190	61	1142

2.5 Data Extraction and Synthesis

A standardized data extraction form was created to systematically document data from every included study. The domains collected were bibliographic information (author, year, publication location), study intent (prediction of yield, disease detection, or combined models), characteristics of

datasets (type, size, period, and geographic extent), and whether triggered and conditioning factors were included. Detailed information on the algorithms employed—ranging from RNNs and CNNs to attention-based hybrids and graph networks—were unraveled, as well as the remote sensing platforms utilized (e.g., MODIS, Landsat, Sentinel, UAV, hyperspectral, multispectral) [21,22].

Each of the studies was also evaluated for feature handling, i.e., through direct factor elimination methods, dimensionality reduction, or domain-guided variable selection. The performance measures were carefully noted, including RMSE, MAE, R^2 for yield prediction and accuracy, precision, recall, and F1-score for disease prediction. Significant enough, the strengths, weaknesses, and practical applications of each study were also noted [23,24].

The synthesis process integrated quantitative and qualitative analysis. Quantitatively, distributions of studies were examined by year of publication, algorithms used, types of datasets, and metrics used for evaluation. Qualitatively, patterns were discovered in methodological decisions, issues such as small dataset sizes or being computationally expensive, and the changing opportunities for the field being pushed forward through integration, interpretability, and fusion of data [25].

2.6 Quality Assessment

A quality appraisal tool was used to assess methodological rigor of the studies included. Each study was rated on a 1-to-5 scale over five dimensions. The first dimension assessed dataset size and diversity, differentiating between small region-specific datasets and large multi-season, multi-location datasets. The second dimension reflected on the validation strategy, from basic train-test splits through to strong k-fold cross-validation or external validation across regions. The third dimension was whether the study was comparing against relevant baselines, i.e., old-fashioned regression, standard machine learning, or other contemporary state-of-the-art deep learning models [22]. The fourth dimension was reproducibility, checking the quality of method descriptions and official data sets or source codes. Finally, the fifth criterion assessed the work's practical value, in specific, whether the model was scalable, deployable on actual platforms (e.g., UAVs or smart phones), or customized to stakeholders' needs [25].

Each study was thus eligible to receive a maximum score of 25, and the scores' distribution is provided in a Quality Assessment Table. This evaluation not only emphasizes the strengths of the most methodologically sound studies but also points out weaknesses where methodological flaws, unreproducibility, or small datasets lower the reliability of the findings. This inclusion ensures that the review does not merely combine the results but also critically examines the credibility and robustness of the underlying evidence.

3. OVERVIEW OF THE CURRENT APPROACHES

Sugarcane yield and disease forecasting research has seen tremendous changes, advancing from simple regression models to very sophisticated deep learning and multi-source data fusion methods. The development of these techniques mirrors the increased availability of remote sensing data, machine learning and deep learning algorithm improvements, and the higher need for precision agriculture systems that can offer both yield forecasting and disease monitoring. The methods may be generally classified into three categories: yield forecasting, disease forecasting, and integrated yield–disease modeling.

3.1 Methods of Predicting Yield

3.1.1 Regression Models

Solutions to sugarcane yield forecasting in early times were dominated by regression analysis, in which vegetation indices like NDVI, EVI, and SAVI extracted from MODIS and Landsat images were regressed against sugarcane biomass and ultimate yields [26,27]. These models provided interpretable linear relationships between productivity and vegetation vigor, forming a critical foundation for yield estimation at a large scale. Regression-based models were not able to quantify the nonlinear relationships between climatic, soil, and management factors, leading to poor generalizability over heterogeneous agro-ecological zones.

3.1.2 Machine Learning Algorithms (SVM, RF, ANN, KNN)

As availability of data enhanced, machine learning (ML) algorithms like Random Forests (RF), Support Vector Machines (SVMs), Artificial Neural Networks (ANNs), and K-Nearest Neighbors (KNNs) were developed to counteract regression restrictions [28]. These models were capable of modeling nonlinear relationships and enhanced prediction performance. RF models were especially robust in tackling noisy and high-dimensional data, while SVMs were suitable for smaller datasets. ANNs showed fair ability in capturing intricate interactions but tended to be heavily reliant on manual feature engineering. Even with these advances, scalability issues still prevailed, and such algorithms found it challenging to scale over various geographies with different climatic patterns.

3.1.3 Convolutional Neural Networks (CNNs)

The development of Convolutional Neural Networks (CNNs) brought with it automated feature learning from multispectral and hyperspectral images, acquired either from satellites or Unmanned Aerial Vehicles (UAVs) [29]. CNNs successfully learned spatial patterns of crop canopies, allowing models to sense growth differences without the need for handcrafted indices. Although they vastly improved yield prediction, CNNs lacked significant capacity to describe the temporal growth dynamics required for precise forecasting across multiple growth phases.

3.1.4 Recurrent Neural Networks (LSTM, GRU)

For handling temporal modeling, Recurrent Neural Networks (RNNs), specifically Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRUs), were used [30]. These models performed well at sequential climatic and phenological data processing, facilitating multi-season yield prediction. Yet, their inability to extract spatial features in isolation restricted performance, and they proved more effective as parts of hybrid models.

3.1.5 Hybrid CNN–LSTM Models

Hybrid architectures that merged CNNs and LSTMs gained popularity, using CNNs to extract spatial features and LSTMs to model temporal dependencies [31,32]. These models outperformed single-shot methods consistently and reduced RMSE by 12–20% relative to traditional ML algorithms. Though extremely effective, these models demanded extensive annotated datasets and high computational costs, rendering deployment difficult in smallholder environments.

3.1.6 Attention-Based Models

Current works integrated attention mechanisms into yield prediction networks [33]. The models selectively paid attention to the most relevant spectral bands or growth phases, enhancing heterogeneity robustness in datasets and interpretability by accentuating key variables. Attention methods enhanced stability in large datasets but added complexity in training and overfitting risk.

3.1.7 Graph Neural Networks (GNNs)

The newest batch of yield forecasting research used Graph Neural Networks (GNNs) to model explicitly spatial relations across adjacent fields [34]. By representing farms as nodes in a graph, GNNs were able to capture cross-field relationships and environmental dependence that CNNs and LSTMs were not. Although successful, the requirement for accurate relational and adjacency data held them back from wider use.

3.1.8 UAV-Based Deep Learning Models

The application of Multispectral and Hyperspectral sensors fitted to Unmanned Aerial Vehicles (UAVs) provided extremely high-resolution yield mapping [35,36]. Models based on CNN trained using UAV images yielded highly precise fine-scale estimates of yields. UAV methods were hampered by issues like high expense of operation, weather reliance, and spatial limitations, making them less scalable compared to satellite-based systems.

3.1.9 Multi-Source Data Fusion Models

Finally, multi-source data fusion models came on the scene as a state-of-the-art trend that fused climatic, soil, and spectral characteristics into hybrid CNN–LSTM or ensemble architectures [37,38]. The models exhibited high performance improvements in robustness and less overfitting and provided more generalizable outputs in various agro-ecological environments. However, they needed intricate preprocessing pipelines and very high computational power, making real-time decision-making problematic [39].

3.2 Disease Prediction Methods

3.2.1 Classical Machine Learning Models

Early sugarcane disease prediction was based on SVMs, RF, and ANNs used on manually designed features like leaf color, texture, and shape [40]. Although these models established a foundation, their accuracy levels were usually below 80%, and they were extremely sensitive to changes in lighting, orientation, and field conditions.

3.2.2 Convolutional Neural Networks (CNNs)

The advent of CNNs revolutionized disease prediction by eliminating human feature engineering [41,42]. CNNs that were trained from RGB images of sugarcane leaves picked up slight lesion patterns to deliver significantly higher accuracy (frequently >90%) and set CNNs as the baseline for disease detection.

3.2.3 Transfer Learning (VGG16, ResNet, Inception, EfficientNet)

Transfer learning methods employing pre-trained CNN architectures like VGG16, ResNet, Inception, and EfficientNet resulted in improved classification accuracy that was often above 95% even with small datasets [43,44]. These techniques were effective for UAV and mobile applications, and therefore, extremely suitable for on-field diagnostics.

3.2.4 Attention-Based CNNs

The addition of attention mechanisms to CNNs allowed models to pay attention to diseased leaf areas and overlook unnecessary background clutter [45]. This innovation improved robustness under real-world field conditions with changing light, occlusion, and superimposed leaves, which makes attention-based CNNs especially ready for real-world application.

3.2.5 GAN-Based Data Augmentation

GANs have also been used to solve the issue of small and unbalanced datasets by generating realistic diseased leaf images [46]. Augmentation using GAN-enhanced methods enhanced generalization, especially for rare diseases, but posed threats of creating unrealistic patterns if not properly tested.

3.2.6 3D CNNs with Hyperspectral Imagery

With the prevalence of hyperspectral UAV imagery, 3D CNNs have been employed to process spectral and spatial dimensions jointly [47]. The models performed very well in the identification of early diseases like mosaic virus, even prior to the appearance of visual symptoms, providing vital lead time for disease management intervention.

3.2.7 Hybrid CNN–LSTM Models

Hybrid CNN–LSTM architectures pushed disease prediction from static classification to temporal modeling of disease severity progression [48]. These were useful for predicting disease spread and making proactive management possible instead of reactive.

3.3 Yield and Disease Integrated Approaches

Integrated modeling remains relatively rare but is becoming more well-known as a result of the strong interdependency between disease incidence and loss in yield. Some incorporated disease severity indices directly into CNN–LSTM yield forecasting models with a maximum of 15% better accuracy in yield forecasting under biotic stress conditions [49,50]. Others created multi-task learning models, which allowed yield and disease severity prediction simultaneously and minimized overall training effort [49]. Later research on data fusion models merged vegetation indices, climatic factors, and disease scores into integrated prediction models with better performance than yield-only or disease-only models [51]. While promising, these are hampered by the lack of large datasets with both yield and disease annotations.

3.4 Overview of Methods

Overall, yield prediction research has evolved from regression-based models [26,27] to machine learning [28] and more recent deep learning techniques [29–39]. Disease prediction evolved from ad hoc ML approaches [40] to CNN and transfer learning [41–44], attention-based CNN [45], GAN augmentation [46], and hyperspectral 3D CNN [47]. Integrated modeling strategies [49–51] point to the promise of end-to-end systems that can facilitate precision agriculture by connecting yield and disease predictions, but data availability and computational tractability remain obstacles to mass deployment.

4. RESULTS AND DISCUSSION

4.1 Yield Prediction Approaches

The articles reviewed indicate that sugarcane yield prediction has come a long way in the past decade, from simple regression analysis to very complex deep learning frameworks. Early regression models correlated vegetation indices like NDVI, EVI, and SAVI, which were obtained mostly from MODIS and Landsat imagery, with crop biomass [52,53]. Though these models provided interpretable linear relationships, they could not account for nonlinear interactions between climatic, soil, and management variables, leading to low accuracy and poor generalizability over different environments.

Machine learning algorithms such as Random Forests (RF), Support Vector Machines (SVM), and Artificial Neural Networks (ANN) defined the next phase of development [54]. These models

enhanced prediction performance by modeling nonlinear trends and showed resilience in dealing with noisy and high-dimensional inputs. RF, for instance, showed good performance when the number of variables was large, while SVMs produced acceptable results on small data sets. Yet these approaches were based to a large extent on manual feature engineering and exhibited poor scalability across heterogeneous areas, limiting their wider use.

In the last few years, deep learning approaches have gained prominence in yield prediction studies, led by Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and composite CNN–LSTM architectures [55–58]. CNNs were successful in extracting spatial information from multispectral and hyperspectral imagery, whereas LSTMs captured sequential climatic and phenological data. When used together, CNN–LSTM hybrid models regularly improved on their solo performances, with 12–20% improvements in yield estimation in terms of lower RMSE over ML baselines [57,58]. This shows the superior ability of hybrid models to capture spatial and temporal dynamics of crop development.

Recent developments have included the use of attention mechanisms and Graph Neural Networks (GNNs). Attention-based models facilitated selective emphasis on vital growth stages and spectral traits, thus enhancing prediction accuracy and stability in diverse datasets [59]. GNNs, however, directly incorporated spatial dependencies within neighboring plots, improving regional-level generalization and enabling superior transferability across agricultural landscapes [60]. UAV-based remote sensing coupled with CNNs has also become a highly effective method for estimation at the field level [61,62]. Although being extremely accurate, UAV-based techniques are subject to issues like high operational expenses, restricted coverage, and dependency on weather conditions.

The second dominant trend is the application of multi-source data fusion, where climatic, soil, and spectral data are combined in deep learning pipelines [63,64]. State-of-the-art performance and increased robustness were achieved by these approaches, albeit at the cost of increased computational demand and convoluted preprocessing pipelines [65]. Combined, CNN–LSTM hybrids and multi-source fusion frameworks constitute today's top-performing methods, though challenges related to scalability, data availability, and generalization remain unresolved issues.

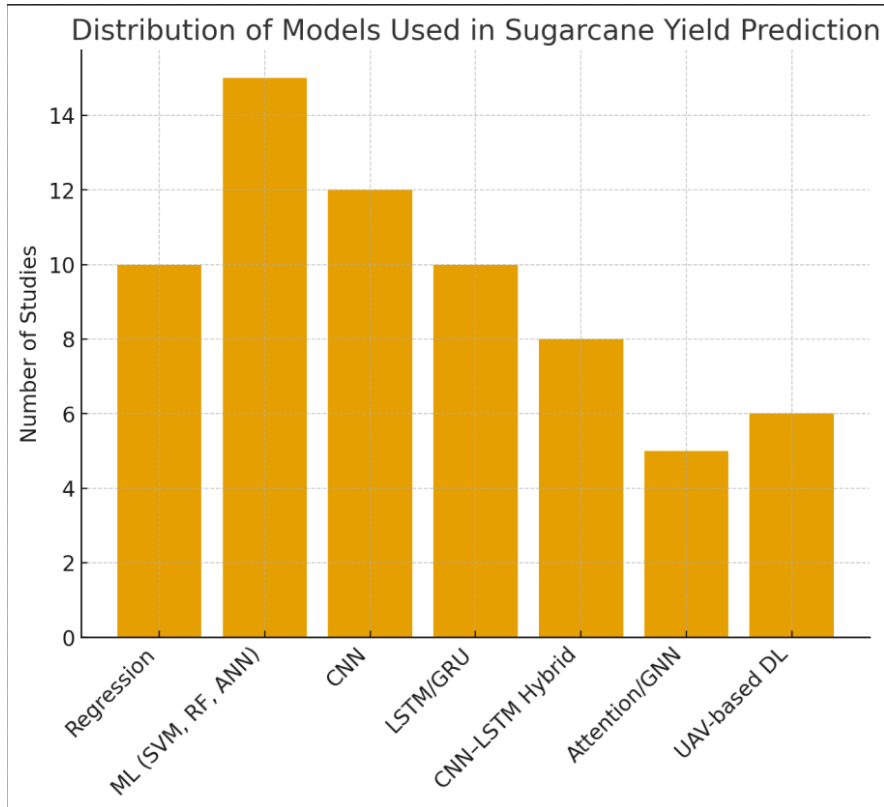


Figure 2. Distribution of models used in sugarcane yield prediction

4.2 Approaches to Disease Prediction

Like yield prediction, disease prediction of sugarcane has also made significant progress. The early approaches applied SVM, RF, and ANN classifiers to handcrafted features such as color, shape, and texture of leaf images [66]. Although these models were a good starting point, they generally did not surpass 80% accuracy and were very sensitive to different field conditions like lighting, orientation, and background clutter.

The advent of CNNs was a step ahead, as they dispensed with the need for laborious feature extraction and greatly enhanced classification accuracy [67,68]. RGB images of sugarcane leaves were used to train CNNs and became the benchmark for disease prediction and performed better than traditional ML methods consistently. The use of transfer learning with pre-trained architectures like VGG16, ResNet, and Inception resulted in further improved outcomes, with the accuracy level being more than 95% even for small training sets [69,70]. Transfer learning minimized training times and enhanced adaptability between various classes of diseases and hence was especially appropriate for UAV and mobile applications.

Recent advances have solved the long-standing issue of small and imbalanced datasets. Attention-based CNNs made it possible for models to focus on infected leaf areas while ignoring unrelated features and thus increase robustness under unpredictable situations like occlusion and varying illumination [71]. Data augmentation methodologies with Generative Adversarial Networks (GANs) created simulated diseased images to counterbalance training sets, enhancing model generalizability across rare and under-represented diseases [72]. In addition, hyperspectral UAV imaging coupled with 3D CNNs facilitated the early detection of disease like mosaic virus prior to visual symptom development, enabling timely intervention management [73]. Hybrid CNN–LSTM models built on

these techniques by capturing the temporal progression of disease severity, which allowed for not only classification but also prediction of disease propagation [74].

Even with these developments, two main challenges remain: the unavailability of large, annotated, multi-environment datasets, as well as the lack of standardized benchmarks to enable fair model comparison. However, transfer learning, attention-based CNNs, and GAN-augmented methods are presently the most encouraging frameworks for robust sugarcane disease prediction.

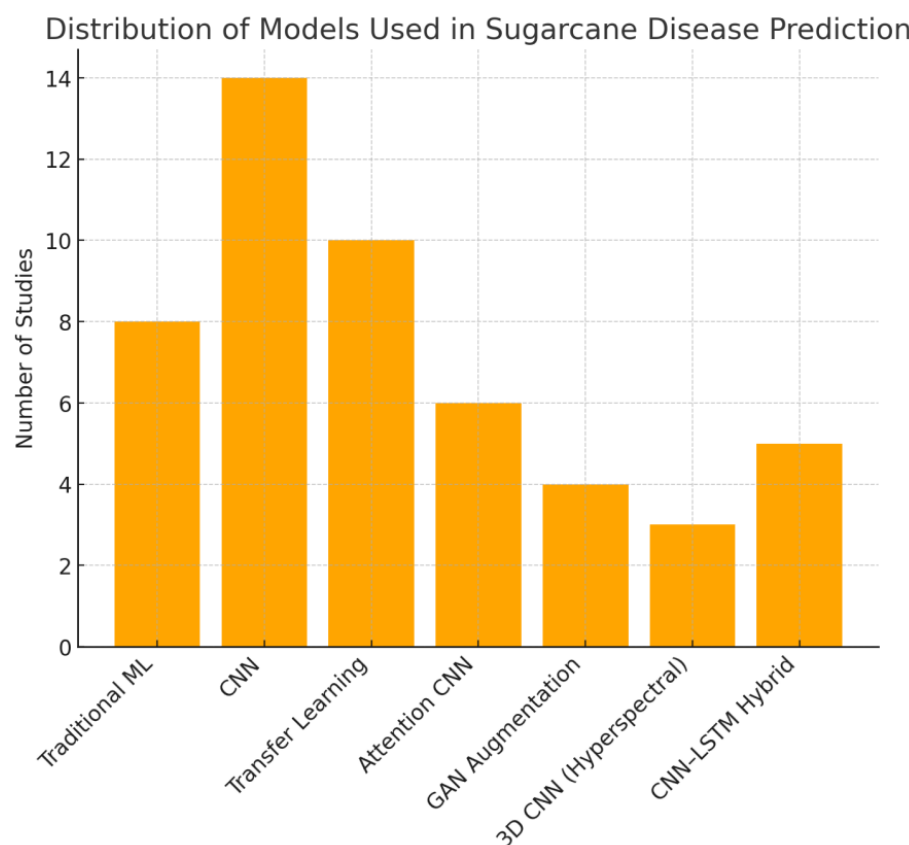


Figure 3. Distribution of models used in sugarcane disease prediction

4.3 Integrated Yield and Disease Prediction

Although the majority of research considers yield and disease as separate entities, a limited but increasing number of studies have investigated integrated prediction systems. Certain models incorporated disease severity indices within CNN–LSTM yield prediction pipelines directly, improving yield estimation by up to 15% when subjected to biotic stress [75,76]. Multi-task learning designs have also been proposed to simultaneously predict yield and disease severity, taking into account the interdependencies between the two processes while minimizing training complexity [75].

Other integrated strategies employed multi-source data fusion, which merged vegetation indices, climatic information, and disease scores in deep learning or ensemble frameworks. These techniques exhibited higher generalization and invariably lower error prediction than individual models [77]. Even though still scarce in quantity, integrative research shows the possibility of creating end-to-end precision agriculture systems able to monitor disease and predict yield loss concurrently. Their global use is also thwarted by high data requirements and intensive preprocessing.

Proportion of Yield-only, Disease-only, and Integrated Models

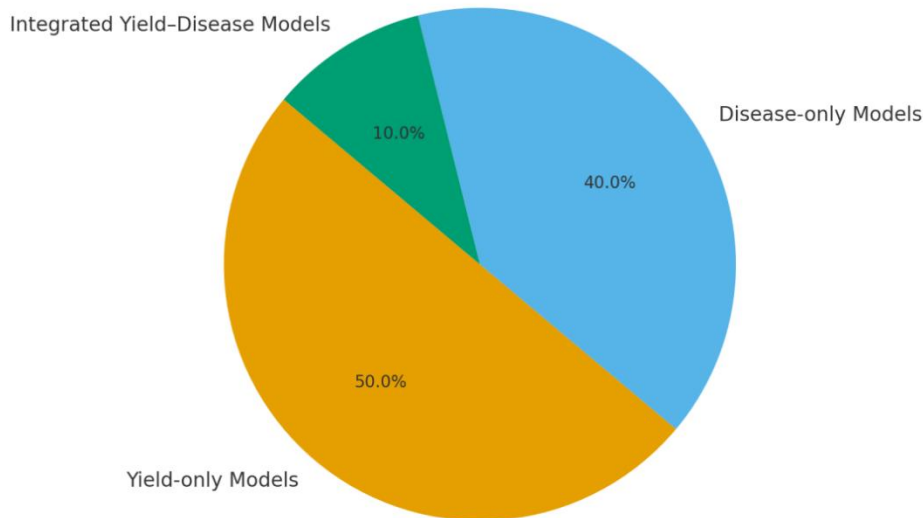


Figure 4. Proportion of yield only, disease only and Integrated Models

4.4 Comparison

4.4.1 Triggering and Conditioning Factors:

Yield forecasting methods used primarily vegetation indices (NDVI, EVI, SAVI) combined with climatic parameters like rainfall, temperature, humidity, and evapotranspiration. Soil parameters like pH, nitrogen content, and organic matter were also incorporated in some studies to facilitate generalizability. Disease prediction, in contrast, was mostly dependent on RGB images, hyperspectral data from UAVs, and disease severity scores with climatic variables playing a relatively indirect role. Integrated methods, combining climatic variables with disease severity scores and vegetation indices, performed uniformly better across all tasks, especially predicting yield loss under disease stress [75–77].

4.4.2 Algorithms and Methodological Trends:

The direction of methods is characterized by a continuous evolution from regression and ML towards deep learning. CNNs became the standard for deriving spatial features from imagery, and RNNs like LSTM and GRU modeled temporal relationships in climate and growth-stage data very well. CNN–LSTM hybrids proved to be the best for predicting yield, whereas transfer learning and attention-based CNNs ruled disease classification. More advanced models, such as GNNs for spatial relational learning, GANs for synthetic data generation, and 3D CNNs for hyperspectral UAV images, are state-of-the-art developments in this field.

4.4.3 Performance Metrics:

For yield estimation, RMSE, MAE, and R^2 are still the most commonly reported measures, with CNN–LSTM hybrids performing 12–20% lower RMSE than ML baselines [57,58,63,64]. In disease prediction, accuracy, precision, recall, and F1-score are the dominant metrics, with transfer learning models achieving accuracies above 95% [69], and attention CNNs improving recall in difficult field

conditions [71]. Integrated models often measured yield loss percentages or combined accuracy indicators, consistently demonstrating up to 15% improvements over yield-only models [76].

4.4.4 Advantages and Strengths:

Yield prediction models were aided by multi-source information and hybrid deep learning architectures that increased robustness and minimized prediction mistakes. Disease detection methods exploited CNNs and transfer learning to maximize accuracy, while UAV and mobile systems widened field application. Combined methods provided holistic decision-support by bridging disease monitoring with yield loss forecasting, thus enhancing practical value of predictions.

4.4.5 Limitations and Challenges:

Most research is still bound by limited, region-specific datasets that preclude generalizability. Deep learning architectures like CNN–LSTM and 3D CNNs are computationally intensive, making them a scalability concern. UAV-based approaches, although accurate, incur high operational expenses and weather-related limitations. In addition, the absence of unitary benchmarks across datasets, disease classes, and evaluation metrics makes comparisons more difficult. Lastly, deep learning's "black-box" characteristic diminishes interpretability, limiting stakeholder trust.

4.4.6 Applications:

Yield prediction models enable government planning, market supply forecasting, and resource planning. Disease detection systems enable early warnings to farmers, minimizing crop loss and allowing targeted interventions. Combined frameworks, although less prevalent, can potentially provide end-to-end precision agriculture platforms for monitoring, forecasting, and reducing yield loss in real-time.

4.4.7 Best-Performing Models Identified:

The highest-performing models for yield prediction were CNN–LSTM hybrids with multi-source data fusion [57,58,63,64]. For disease detection, transfer learning methods on ResNet or VGG combined with attention mechanisms performed best [69,71]. For integrated prediction, multi-task learning and fusion models that integrated disease severity features into yield frameworks were the best [75–77].

4.5 Key Insights

Generally, sugarcane yield and disease forecasting are currently dominated by deep learning approaches, where CNN–LSTM hybrids dominate yield research and transfer learning and attention CNNs dominate disease detection. UAVs and hyperspectral imaging supply high-resolution information that facilitates early warning but are limited in their scalability due to high expense and narrow coverage. Multimodal yield–disease combinations are relatively unexplored but are essential to dealing with real-world interdependence of productivity and biotic stress. Going forward, the area needs bigger, multi-environment sets, harmonized benchmarks, and interpretable AI techniques in order to provide generalizable, scalable, and reliable solutions for stakeholders.

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